CHRONOS RESEARCH PROJECT

WORKING PAPER 01

The Anatomy of a Polymarket Price

Why prediction-market probabilities move in sticky jumps, not smooth diffusion

A mathematical and machine-learning study of 803,862 hourly observations

62.4%	157.1	112/113	0
unchanged	**kurtosis**	**event comparisons**	**trade-ready signals**
interior hourly samples	Gaussian benchmark: 3	full model beat state-only	after validation and costs

The answer in one paragraph

Polymarket probabilities do not move like a stock price following a smooth random walk. Most recorded hours show no change at all. When the probability does move, the change is often abrupt, its size depends on recent movement, and its direction is more likely to reverse than to continue. Most importantly, a market that has been quiet for a long time is much less likely to update in the next hour than a market that has just moved. A model that treats these as three separate questions — *will it move, in which direction, and by how much?* — fits later data materially better than a bounded Gaussian diffusion. The finding is statistically useful, but it is not yet a trading strategy: the strongest model predicts activity, not profitable direction.

What we studied

Recorded YES probabilities from resolved Polymarket contracts, grouped by event and ordered in time. A second cohort and official five-minute data were used to challenge the first result.

What we found

A jump process that explicitly allows no change, and remembers inactivity and recent movements, describes the recorded data substantially better than smooth diffusion.

What we did not find

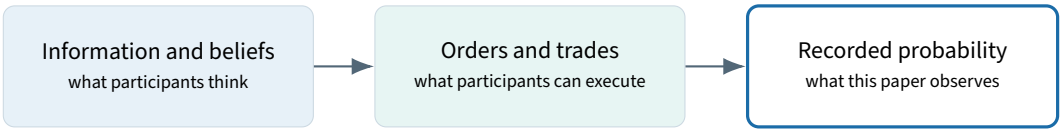
No protected, executable trading edge. Apparent flow and external-data signals weakened or became negative after direction, fees, liquidity, and temporal controls.

Status. Development evidence, unchanged later-cohort replay, and one retained protected negative. Independent research; not affiliated with Polymarket. Not investment advice.

1. What exactly is a Polymarket price?

Suppose a contract asks: *"Will Candidate A win the election?"* A YES share pays \$1 if the answer is yes and \$0 if the answer is no. If a YES share trades around \$0.65, people often say that the market assigns a 65% probability to the event.

That interpretation can be useful, but it is not automatic. The displayed number is produced by orders, trades, market rules, and a data feed. It may differ from the probability in traders' heads, from the best price available for a new order, and from the true chance that the event will happen. Prediction-market theory makes those distinctions explicit [8, 6].



The paper studies the final box. It does not assume that a recorded probability is the true probability, an executable quote, or a direct measurement of collective belief. Those are separate empirical questions.

Why ordinary stock-price intuition is incomplete

An equity can, in principle, keep rising for decades. A binary contract cannot. Its price lives between 0 and 1 and eventually resolves to exactly one of those values. News may arrive in bursts, and long periods may pass with no meaningful update. There is also no continuously traded underlying asset that lets an investor replicate the contract in the Black–Scholes sense.

This changes the mathematical starting point:

Question	Conventional default	Prediction-market complication
State space	Often unbounded or positive	Probability is bounded in $[0, 1]$
End point	Asset continues trading	Contract resolves to 0 or 1
Information clock	Often approximated as continuous	News and participation arrive irregularly
Replication	Underlying asset may be traded	Usually no replicating underlying
Observation	Trade or quote has a defined role	Sample may mix feed, quote, and trade mechanics

The two questions

1. Which parts of financial time-series theory remain useful once these structural differences are respected?
2. Which stochastic-process family best describes and predicts the recorded Polymarket probability series?

The first paper deliberately stops before claiming a strategy. A process can be predictable in *timing* or *volatility* while remaining useless for predicting profitable direction.

2. Data designed to survive skeptical review

The main analysis uses 562 resolved markets from 157 event groups. It contains 803,862 recorded hourly probabilities. A later cohort contributes 29 new event groups, and ten preselected markets are checked against independently acquired official five-minute price histories.



Evidence set	Events	Markets	Observations	Purpose
Main development cohort	157	562	803,862	Description and chronological model fitting
Later cohort	29	147	42,549[a]	Replay the frozen law without refitting
Official five-minute panel	10	10	19,258	Test whether hourly sampling creates the result
Formal arrival schedule	111	–	231 decisions	Governed model evaluation

[a] Observations used in the later process-signature replay. Exact corpus identities appear in Appendix B.

Four rules prevent an attractive story from becoming a false result

1. Time only moves forward
A feature may use only information already available at the forecast time. Training events must occur before test events.

2. Events, not rows, carry the uncertainty
Markets tied to one real-world event are dependent. The analysis gives each event equal total weight and keeps whole events together.

3. Later evidence must be genuinely later
The strongest law is fitted once, frozen, and applied unchanged to a later cohort. This tests transfer rather than another round of tuning.

4. Failure is part of the system
A model that cannot legally produce a forecast has failed that decision. Missing forecasts cannot be removed after their outcomes are seen.

Three clocks are kept separate

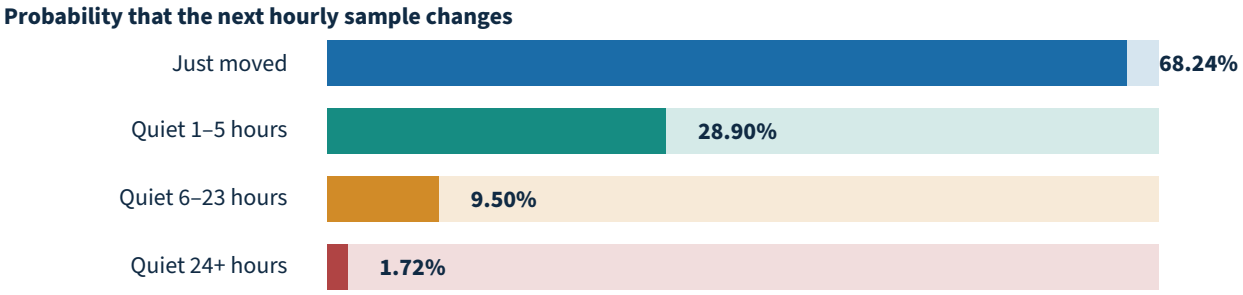
The analysis distinguishes the next recorded sample, the actual elapsed UTC time, and the time remaining until market close. Headline one-hour changes require an elapsed gap between 45 and 75 minutes. A 24-hour variance calculation is accepted only when the elapsed time is close to 24 hours. This avoids treating irregularly spaced observations as if they were equally spaced.

Why this matters. Randomly shuffling individual rows would let nearly identical contracts from the same event appear in both training and test data. It would also let future market regimes inform the past. Both shortcuts can make weak models look impressive.

3. What the probability paths actually look like

The series is dominated by inactivity punctuated by abrupt updates. That is not a metaphor: among regular interior hourly observations, 62.40% are exactly unchanged. The distribution of changes also has far more extreme observations than a Gaussian law: its raw kurtosis is 157.10, compared with 3 for a Gaussian distribution.

The longer a market stays quiet, the less likely it is to move next



A market that has already been unchanged for at least 24 hours moves in the next recorded hour only 1.72% of the time. A market that just moved changes again 68.24% of the time. A constant-rate Poisson clock cannot explain a difference of this size. In statistical language, the arrival hazard has memory; in plain language, activity clusters.

Five signatures contradict smooth diffusion

Observed property	Main estimate	Why it matters
Exact zero changes	62.40% interior hourly	A continuous Gaussian law has no point mass at zero
Extreme tails	Kurtosis 157.10	Large updates occur far more often than Gaussian scaling suggests
Sub-diffusive scaling	24h ratio 0.235 pooled	Variance does not grow linearly with clock time
Sign reversal	62.21% of adjacent nonzero moves	Recorded directions are not independent
Boundary persistence	98.99% low; 97.07% high	Contracts behave differently near 0 and 1

Could the data feed be creating the pattern?

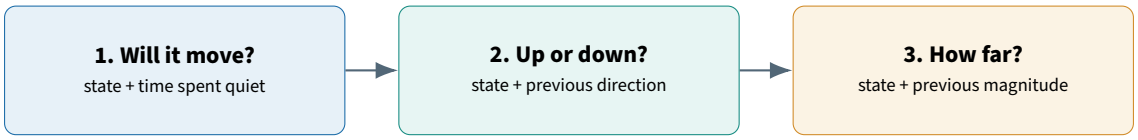
That is the most important rival explanation. The five-minute panel therefore repeats the test on an independently acquired official source. The pattern becomes stronger, not weaker: 78.00% zero changes, kurtosis 203.64, reversal 67.87%, and a 24-hour variance ratio of 0.067. Of 1,580 hourly samples that can be aligned to the official history within ten minutes, 99.49% match exactly.

The test does *not* prove that latent beliefs jump. It shows that coarse hourly sampling alone does not manufacture the observed structure on this panel. The registered series may still mix economic inactivity, stale quotes, rounding, bid/ask effects, and the way the venue publishes its history.

Finding 1. The recorded probability process is sticky and bursty. A useful model needs an explicit probability of no change, not merely a small continuous volatility.

4. A model built around how the market updates

Rather than force every hour into one return distribution, the leading model asks three questions in sequence:



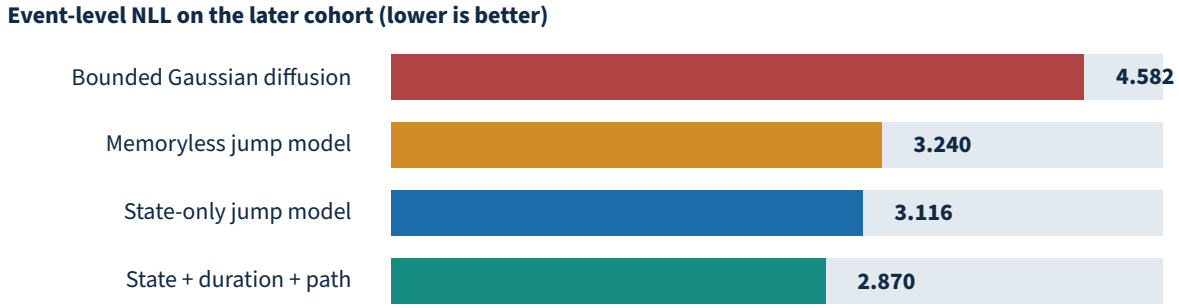
Let P_t be the recorded YES probability. The model first assigns probability q_t to a nonzero change. Conditional on moving, it assigns probability s_t to an upward move and models the log size of the change with a Gaussian distribution:

$$\begin{aligned} \Pr(\Delta P_{t+1} = 0 \mid \mathcal{F}_t) &= 1 - q_t, \\ \Pr(\Delta P_{t+1} > 0 \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t) &= s_t, \\ \log|\Delta P_{t+1}| \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t &\sim \mathcal{N}(\mu_t, \tau_t^2). \end{aligned}$$

The information set $\mathcal{F}_t$ contains only legally available features: current probability, time to close, quiet duration, recent inactivity, previous nonzero direction, and previous nonzero size. Appendix A gives the exact scoring rule.

The full model wins because each layer contributes

The models are judged by negative log likelihood (NLL): how much probability they assigned to what actually happened. Lower is better. Event-level NLL first averages within each event and then gives every event equal weight.



In the development cohort, the full model improves event-level NLL by 7.74% over the state-only jump model and wins 112 of 113 event comparisons. The model is then frozen and applied, without refitting, to the later cohort. It improves by 7.88% and wins all 27 eligible events against both the state-only model and diffusion.

Absolute scores are worse in the later cohort. That matters: the mathematical structure transfers, but the parameter levels change. The result supports a stable *family* of models with regime-dependent parameters, not one universal coefficient vector.

Finding 2. A bounded, zero-inflated marked-jump model with duration and path information describes later Polymarket data better than smooth diffusion. The result transfers without refitting, although its exact calibration changes across cohorts.

5. Strong predictive component, rejected protected system

The most reliable part of the model is the first question: *will the recorded probability change in the next hour?* A regularized logistic model uses current state, time to close, the logarithm of the quiet run, and the fraction of unchanged observations during the previous 24 hours.

LATER COHORT: EVENT-LEVEL BRIER SCORE

0.261 → 0.174

State only + inactivity

33.38% improvement; 27 wins in 27 eligible events.

WHAT A BRIER SCORE MEANS

It is the average squared error of a probability forecast. Zero is perfect. The model's mean forecast was 41.29%; the next hour changed 45.34% of the time. The absolute calibration gap was 4.05 percentage points.

This is meaningful predictive evidence. It is also narrower than it first appears. The model says when a recorded update is likely; it does not say whether the update will be up or down, whether an order can be filled, or whether a trade will earn more than its costs.

Why the final system was rejected

After development, an exact version of the continuous inactivity model was frozen: source code, features, training window, decision schedule, metrics, and failure rules. It was then evaluated once on protected data.



On the 65 completed decisions, Brier score was 0.189 and mean absolute error was 0.394, both inside the predeclared limits. But the system was required to complete at least 95 of 100 decisions. It completed only 65. Thirty-four decisions lacked the fixed legal training history and one lacked a legally available outcome. Error concentration also narrowly missed its limit.

Protected decision: rejected. Coverage was 65% against a required 95%; failure rate was 35% against a maximum 5%. The completed rows cannot be presented as a successful model because the missing decisions may be systematically harder than the completed ones.

This distinction is central. A model is not merely an equation evaluated on convenient rows. It is the complete decision process, including whether the required history exists and whether it can produce forecasts on schedule. The protected failure does not erase the earlier evidence that inactivity matters. It prevents that evidence from being promoted into a broader validated claim.

Finding 3. Quiet duration is a powerful predictor of recorded update timing. The exact protected implementation is nevertheless rejected because it failed too many declared decisions.

6. What this changes - and what comes next

The evidence supports a different default model for Polymarket research. Instead of beginning with a smooth return process and adding occasional anomalies, begin with an event-linked process that can remain unchanged, update in clusters, and terminate at a binary boundary.

Financial idea	Verdict	Required modification
Chronological validation	Transfers	Split and weight by real-world event, not individual rows
State-dependent volatility	Transfers	Respect probability bounds and terminal resolution
Duration and point-process models	Transfers strongly	Model time since the last update as part of the state
Heavy-tailed returns	Transfers conditionally	First separate zero changes and conditional size
Martingale reasoning	Useful benchmark	Distinguish sampled price, executable price, belief, and physical probability
Constant Brownian diffusion	Poor default	Add point mass, jumps, irregular information time, and boundaries
Black–Scholes replication	Does not transfer directly	No standard replicating underlying or complete market

Why this is not yet a trading paper

The strongest result predicts movement timing. Directional ideas have been tested separately and have not survived the full controls:

- A raw 15-minute flow association near -0.46 fell to about $+0.04$ to $+0.05$ after transaction direction was handled correctly.
- In the Brazil election study, 31 poll releases produced 25 usable decisions and 17,570 canonical fills. The broad 72-hour fade earned \$10.69 before costs and lost \$18.97 after conservative fees.
- Other event and internal-market rules were also rejected or remained descriptive because they lacked executable quotes, sufficient breadth, or protected confirmation.

These are useful results. They show why statistical association, price prediction, and tradable edge are three different standards.

The next research programme

1. Use the inactivity hazard as an *eligibility layer*: predict direction only when a movement is likely enough to make the forecast operationally relevant.
2. Add time-legal external information — including polls, macro releases, and prices from other prediction venues — as predictors, not as presumed arbitrage.
3. Evaluate direction and magnitude chronologically, then translate predictions into side-specific Polymarket orders with spread, fees, latency, depth, and abstention.
4. Reserve a genuinely new protected cohort for any system that clears fixed development gates. The exposed protected schedule from this paper cannot be reused.

Bottom line
Polymarket's recorded probabilities are best understood as a bounded, state-dependent jump process with long quiet periods and clustered updates. That finding is broad, reproducible, and mathematically useful. It is a foundation for better forecasting and simulation — not evidence that a profitable strategy has already been found.

The appendix records the exact equations, corpus identities, robustness checks, negative results, limitations, and references.

Appendix A Exact model and scoring definitions

Let $P_t \in [0,1]$ be the registered YES probability and $\Delta P_{t+1} = P_{t+1} - P_t$. The bounded Gaussian comparator is

$$\Delta P_{t+1} \mid \mathcal{F}_t \sim \mathcal{N}\big(0, \sigma^2(c_t) P_t(1-P_t)\big)\,, \tag{1}$$

where c_t is a fixed probability and time-to-close cell. The bounded factor shrinks local variance near 0 and 1 but retains a continuous Gaussian increment law.

The full hurdle law is

$$\Pr(\Delta P_{t+1} = 0 \mid \mathcal{F}_t) = 1 - q(\mathcal{F}_t), \tag{2}$$

$$\Pr(\Delta P_{t+1} > 0 \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t) = s(\mathcal{F}_t), \tag{3}$$

$$\log|\Delta P_{t+1}| \mid \Delta P_{t+1} \neq 0, \mathcal{F}_t \sim \mathcal{N}(\mu(\mathcal{F}_t), \tau^2(\mathcal{F}_t)). \tag{4}$$

Arrival uses probability and quiet-duration cells; sign uses probability and previous nonzero sign; magnitude uses probability and previous nonzero magnitude. Each cell shrinks toward the event-equal global law with one event of prior weight.

For nonzero x, the model assigns probability to the same 0.001-wide interval for every family:

$$q(\mathcal{F}_t)\Pr(\operatorname{sign} x \mid \mathcal{F}_t)\times \tag{5}$$

$$\left[\Phi\left(\frac{\log(|x|+\epsilon)-\mu}{\tau}\right) - \Phi\left(\frac{\log(\max(10^{-6}, |x|-\epsilon))-\mu}{\tau}\right)\right], \tag{6}$$

with $\epsilon = 0.0005$. A zero receives point mass $1 - q(\mathcal{F}_t)$. Negative log interval probability is averaged by observation and separately by event.

The continuous arrival model predicts $Y_{t+1} = 1[\Delta P_{t+1} \neq 0]$:

$$\operatorname{logit}\Pr(Y_{t+1}=1 \mid \mathcal{F}_t) = \beta_0 + \beta_p f(P_t) + \beta_T f(T_t) + \beta_q \log(1+Q_t) + \beta_z Z_{t,24}, \tag{7}$$

where Q_t is the current quiet run and $Z_{t,24}$ is the previous-24-hour zero fraction. Its Brier score is $n^{-1}\sum_i(\widehat{q}_i - Y_i)^2$.

Appendix B Data authority and corpus identity

The opening 1,000-contract corpus contains 235 events and 1,180,437 observations. The successor acquisition contains 1,509 contracts, 321 events, and 1,585,740 observations. Its content identity is:

```
polymarket-corpus:184b83f7f8012da4ca9116c8b6bad2076570ad5f0bada95b22615ee2f5cceae8
```

The successor partition contains 29 wholly fresh development events (147 markets), 227 embargoed ancestor events (1,005 markets), and 65 protected events (357 markets). Data were acquired through the read-only `human_ro` role. Chronos retains source digests, selection, temporal semantics, partitions, actual field accesses, worker outcomes, and result receipts.

Appendix C Complete numerical results

Model	Development observation NLL	Development event NLL	Later event NLL
Bounded state-dependent Gaussian diffusion	3.511	3.578	4.582
Memoryless zero-inflated marked jump	1.865	2.214	3.240
State-conditioned zero-inflated marked jump	1.782	2.104	3.116
State + duration/path marked jump	**1.585**	**1.941**	**2.870**

Table 1: Complete observation-law comparison. Lower NLL is better.

Protected check	Result	Required	Decision
Brier on completed rows	0.189254	≤ 0.200000	**PASS**
MAE on completed rows	0.394489	≤ 0.400000	**PASS**
Coverage	0.650000	≥ 0.950000	**FAIL**
Failure rate	0.350000	≤ 0.050000	**FAIL**
Maximum error concentration	0.054852	≤ 0.050000	**FAIL**

Table 2: Single protected evaluation of the frozen continuous arrival system.

Appendix D Representation and structural robustness

Fixed view	Increments	Zero rate	Kurtosis	Event-equal 24h ratio
All probability rows	781,388	69.73%	157.10	0.358
Strict interior probability	519,495	62.76%	103.19	0.376
Strict interior log-odds	519,495	62.76%	104.99	0.403
More than 168 hours to close	701,428	69.40%	151.06	0.338
Zero to 168 hours to close	79,960	72.54%	138.44	0.512

Table 3: Boundary, coordinate, and maturity checks.

Whole-event stratum	Zero mass	Variance ratio	Reversal	Quiet ratio
One-market events	0.577	0.403	0.588	0.084
At least five markets	0.527	0.267	0.598	0.032
Lifetime ≤ 30 days	0.477	0.293	0.581	0.038
Lifetime 31–180 days	0.605	0.425	0.594	0.078
Lifetime > 180 days	0.623	0.228	0.643	0.035
Fewer than 500 points	0.458	0.315	0.556	0.014
500–1,999 points	0.586	0.394	0.601	0.070
At least 2,000 points	0.631	0.248	0.635	0.053

Table 4: Event-equal heterogeneity. Structural directions persist; parameter levels change.

Appendix E Negative directional and external-data evidence

Fill-semantics correction

Polymarket's retained fill table is a per-party ledger. Counting maker and taker legs as separate market trades duplicates economic activity and can invert direction. Canonical studies retain the transaction-side record and convert YES and NO activity to a consistent YES-equivalent price and taker direction. The raw 15-minute flow association near -0.46 fell to about $+0.04$ to $+0.05$ after this correction.

Brazil polling study

Thirty-one national Lula/Flávio runoff poll releases were reconstructed from immutable publication history. They yielded 25 decisions and 17,570 canonical Polymarket fills. Features included latest poll change, rolling-three median change, surprise against the previous five releases, and same-pollster change. Follow and fade rules were tested at 0, 2, and 4 percentage points thresholds over 24, 72, and 168 hours. Every rule with at least five executable pairs was negative after conservative fees. The broad 72-hour fade moved from \$10.69 gross to $-\$18.97$ net over seven trades.

Appendix F Claim ledger and limitations

Claim	Status	Boundary
Smooth constant-intensity diffusion describes registered samples	Rejected	Zero mass, tails, scaling, and duration dependence
State-duration/path jump law improves predictive density	Strong development + later replay	Not a protected finding; targets samples
Inactivity improves update prediction	Strong development + later replay	Exact protected system rejected
Boundaries alone explain stickiness	Rejected as sole cause	Interior and log-odds checks agree
Hourly sampling alone creates the signatures	Rejected on selected panel	Official five-minute source agrees
Sign reversal is executable mean reversion	Not established	Quotes, spread, fees, and latency absent
Chronos has found tradable alpha	No	Cost-aware directional ideas rejected

- Registered samples are not identified as trades, quotes, midpoints, latent beliefs, or executable prices.
- The five-minute panel contains ten preselected markets; it is not a venue-wide quote study.
- Event grouping mitigates but cannot remove every dependence among related contracts.
- The diffusion comparator is intentionally simple and does not eliminate every latent continuous-time model with microstructure noise or endogenous sampling.
- One protected system was evaluated and rejected. Its exposed holdout cannot be reused.
- No result establishes causal information arrival, objective-probability equality, Sharpe ratio, or net profit after all execution costs.

Appendix G References and reproducibility

[1] Bacry, E., Delattre, S., Hoffmann, M., and Muzy, J.-F. (2013). Modelling microstructure noise with mutually exciting point processes. *Quantitative Finance*, 13(1), 65–77.

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[5] Majumder, S. R. et al. (2009). Price dynamics in political prediction markets. *PNAS*, 106(3), 679–684.

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[9] Wolfers, J. and Zitzewitz, E. (2006). Interpreting prediction market prices as probabilities. NBER Working Paper 12200.

Numerical claims map to content-addressed receipts in `paper/evidence-index.md`. Replay commands and the non-rerunnable protected boundary are documented in `paper/reproducibility.md`. The manuscript is a presentation layer; it cannot promote the epistemic status of any result.